

RCC Structural Drawing Compliance Report

Project: Foundations for Entrance Gate and Guard Room

Site Location: Kutch, Gujarat (Updated from Malpe Fishery Harbour, Udupi)

Reference Codes: IS 456:2000, SP 34

Step 0: Initial Document Check

- **0.1:** The document is an RCC structural drawing of "FOUNDATIONS" only.
- **0.2:** Site Location: Kutch, Gujarat (Updated based on user input).
- **0.3:** All compliance checks are based only on IS 456:2000 and SP 34.

Step 1: Locate the "NOTES" Section

- **Status:** Compliant

Step 2 & 3: Extract and Verify Design Parameters from "NOTES"

Criteria	Extracted Value	Compliance Check	Status
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1. Grade of Concrete	M25	Updated Analysis: The site location is Kutch, Gujarat. Kutch is known for its arid climate and can experience varying exposure conditions. For foundations, especially in contact with soil, a minimum of M20 is generally required (SP 34, Clause 11.8.4.7). M25 is acceptable for moderate exposure. However, without specific environmental exposure classification (e.g., severe, very severe as per IS 456:2000 Table 5), it's difficult to definitively confirm compliance for all potential conditions in Kutch.	Cannot Verify
2. Reinforcement Bars	HYSD TMT Bars, Grade Fe 550, conforming to IS 1786-1985. Column vertical reinforcement: 8Y16. Column ties: Y8@200. Footing bottom reinforcement: Y10@125.	Updated Analysis: The specified grade is Fe 550. The user input "take FY 1550" seems to be a typo and is interpreted as Fe 550, consistent with the drawing. IS 1786-1985 is an older standard; the current standard for HYSD bars is IS 1786:2008. Using an outdated standard is a non-compliance.	Non-Compliant
3. Lap Length	50 times dia of the bar, not more than 50% of bars are lapped at a section.	Updated Analysis: 50d is a common and generally compliant value for lap length. The staggering requirement is also good practice.	Compliant

4. Clear Cover	Footing/Wall: 50 mm, Columns: 40 mm, Slab: 25 mm, Beam: 30 mm.	Updated Analysis: Footing cover of 50mm is the minimum as per IS 456:2000 Table 16A for moderate exposure. SP 34, Clause 11.8.4.6 specifies 100mm for base slab top, bottom, and sides, and 50mm for column sides. The drawing specifies 50mm for footing/wall, which aligns with IS 456:2000 minimum but is less than SP 34's recommendation for base slabs. Given the 150mm PCC below the footing, the 50mm cover for the footing bottom is acceptable as per IS 456:2000. Column cover of 40mm is compliant (IS 456:2000 Table 16A). Slab and beam covers are also generally compliant for normal exposure.	Compliant (with note on SP 34 recommendation for base slab)
5. Development Length (Ld)	50 times the dia. of the bar.	Updated Analysis: For Fe 550 and M25 concrete, the development length (Ld) can be calculated. From SP 34, Clause 43 (for $f_y = 500 \text{ N/mm}^2$), for M25 concrete, the Ld for tension bars is 48.5d for 10mm bar and 77.7d for 16mm bar. The specified 50d is a general thumb rule. For 10mm bars, 50d is slightly higher than 48.5d, which is conservative. For 16mm bars, 50d is less than 77.7d, which is non-compliant.	Non-Compliant
6. Safe Bearing Capacity (SBC) of soil	10 T/m2 (100 kN/m2). PCC is mentioned (150mm thick).	Updated Analysis: SBC is mentioned. PCC is specified.	Compliant
7. Seismic Zone and Wind Load	Seismic Zone III and Wind Forces as per IS 1893:2016 & IS 875:2015 respectively.	Updated Analysis: Kutch, Gujarat, falls under Seismic Zone V as per IS 1893 (Part 1):2016. The drawing specifies Seismic Zone III, which is incorrect for Kutch. This is a critical non-compliance.	Non-Compliant
8. Building Limitations	Structure is designed for Ground + 1 Storey + Lightweight Roof.	Updated Analysis: Building limitations are specified.	Compliant
9. Structure's Purpose	Entrance Gate and Guard Room.	Updated Analysis: The purpose of the structure is mentioned.	Compliant

10. Floor Heights	Not explicitly mentioned in the "NOTES" section. "First Floor to Terrace Lvl" is indicated in the column detail, but specific floor heights are not listed.	Updated Analysis: User input provides "floor height: 11m". This clarifies the overall height. However, specific individual floor heights (e.g., Ground floor height, First floor height) are still not explicitly detailed in the notes or drawing. SP 34, Clause 3.3.5 states that "heights between floors" should be shown on drawings.	Missing Information
11. Schedule of Footings	Present and consistent with F1 footing details.	Updated Analysis: A "SCHEDULE OF FOOTINGS" table is present and provides dimensions and reinforcement for F1.	Compliant
12. Footing Type	Isolated footings (F1).	Updated Analysis: Isolated footings are used for a Ground + 1 Storey structure, which is appropriate.	Compliant
13. Reinforcement in High-Rise Buildings	Not applicable.	Updated Analysis: The structure is Ground + 1 Storey, not a high-rise building.	Not Applicable
14. Raft Foundation Reinforcement	Not applicable.	Updated Analysis: Raft foundation is not used.	Not Applicable
15. Lift Design	Not applicable.	Updated Analysis: No lift is designed for this structure.	Not Applicable
16. Soil Improvement	"Improved S.B.C of soil considered is 10 T/m ² ". Note 1 in Section X mentions "SIZE STONE LAYERS BELOW THE FOOTING IS TO BE LAID AFTER EXCAVATING AND OMPACTING THE SAME AS PER THE STANDARD PROCEDURE".	Updated Analysis: Soil improvement is indicated by the improved SBC and the mention of stone layers and compaction.	Compliant
17. Column Ties	Column ties are shown in Section X and the C1 detail. Ties are Y8@200. They appear continuous in the C1 detail.	Updated Analysis: Column ties are shown and appear continuous.	Compliant
18. Plan of Ties	A separate plan for ties is not explicitly present, but the C1 column detail shows the arrangement of ties.	Updated Analysis: A dedicated plan for ties is not present, but the detail provides the necessary information for the C1 column. SP 34, Clause 3.3.1 implies that the exact position, shape, size, and spacing of reinforcement should be given. The C1 detail provides this for the column.	Compliant

19. Outer Ties Check	Column vertical reinforcement: 8Y16. Column ties: Y8@200. Footing bottom reinforcement: Y10@125. Percentage of steel in columns: For a 300x450 column, 8Y16 bars (Area = $8 \pi (16/2)^2 = 1608.5 \text{ mm}^2$). Gross area = $300 \times 450 = 135000 \text{ mm}^2$. Percentage = $(1608.5 / 135000) \times 100 = 1.19\%$. This is $\geq 0.8\%$ (IS 456:2000, Cl. 26.5.3.1). Footing reinforcement: Y10@125. For a 2100mm wide footing, area of steel per meter = $(1000/125) \pi (10/2)^2 = 628.32 \text{ mm}^2$. Percentage = $(628.32 / (2100 \times 450)) \times 100 = 0.066\%$. This is less than 0.12% for slabs/footings as per IS 456:2000 (cl. 26.5.2.1).	Updated Analysis: Column steel percentage is compliant. Footing steel percentage is non-compliant as it is less than the minimum required 0.12% for slabs/footings as per IS 456:2000 (cl. 26.5.2.1).	Non-Compliant
20. Cross-Section Area	Not explicitly stated whether gross or effective area is used for calculations in the notes.	Updated Analysis: This information is not explicitly mentioned in the notes.	Missing Information
21. Steel Curtailment	Not explicitly mentioned in the notes.	Updated Analysis: For a G+1 structure, curtailment might not be as critical as in high-rise buildings, but general principles of curtailment should be followed. Not explicitly detailed.	Missing Information
22. Maximum Steel Percentage in Columns	Not explicitly mentioned in the notes. Calculated percentage for C1 is 1.19%, which is well within the 6% (or 4% with lapping) limit (IS 456:2000, Cl. 26.5.3.1).	Updated Analysis: The maximum percentage is not stated in the notes, but the calculated value is compliant.	Missing Information

Step 5: Report Missing or Wrong Information

Grade of Concrete: Cannot definitively verify compliance without specific environmental exposure classification for Kutch, Gujarat, as per IS 456:2000 Table 5. While M25 is generally acceptable, specific coastal or aggressive conditions might require a higher grade.

Reinforcement Bars Standard: The standard for reinforcement bars (IS 1786-1985) is outdated; the current standard is IS 1786:2008. This is a non-compliance.

Development Length (Ld): The specified Ld of 50d is non-compliant for 16mm bars (77.7d required for Fe 550, M25 tension bars as per SP 34, Clause 43).

Seismic Zone: The drawing specifies Seismic Zone III, which is incorrect for Kutch, Gujarat, which falls under Seismic Zone V as per IS 1893 (Part 1):2016. This is a critical non-compliance.

Floor Heights: While the overall height of 11m is provided, specific individual floor heights (e.g., Ground floor height, First floor height) are still not explicitly detailed in the notes or drawing, as recommended by SP 34, Clause 3.3.5.

Footing Reinforcement Percentage: The calculated percentage of steel for footing (0.066%) is less than the minimum required 0.12% for slabs/footings as per IS 456:2000 (cl. 26.5.2.1). This is a non-compliance.

Cross-Section Area: It is not explicitly stated whether gross or effective cross-section area is used for calculations.

Steel Curtailment: Details regarding steel curtailment are not provided.

Maximum Steel Percentage in Columns: The maximum steel percentage allowed in columns is not explicitly stated in the notes.

Summary of Compliance

- **Total Criteria Evaluated:** 22
- **Compliant Items:** 8
- **Non-Compliant Items:** 4 (Reinforcement Bars Standard, Development Length, Seismic Zone, Footing Reinforcement Percentage)
- **Missing Information Items:** 4 (Grade of Concrete verification, Floor Heights, Cross-Section Area, Steel Curtailment, Max Steel Percentage in Columns)
- **Cannot Verify Items:** 1 (Grade of Concrete without exposure class)
- **Not Applicable Items:** 3

Overall Verdict: The integration of new information has highlighted several critical non-compliances and persistent missing information. The change in site location to Kutch, Gujarat, significantly impacts the seismic design requirements, rendering the specified Seismic Zone III non-compliant. The outdated reinforcement bar standard, insufficient development length for larger bars, and critically, the non-compliant minimum reinforcement in footings are major concerns. While some previously missing information (like overall floor height) was provided, specific floor heights and other details remain unaddressed. The number of non-compliant and missing items (8 out of 22 evaluated, excluding N/A) indicates significant areas for revision to ensure full compliance with IS 456:2000 and SP 34.

Recommendations:

Seismic Design Review: Immediately revise the design to comply with Seismic Zone V requirements as per IS 1893 (Part 1):2016 for Kutch, Gujarat.

Reinforcement Standard Update: Specify reinforcement bars conforming to the latest IS 1786:2008 standard.

Footing Reinforcement: Increase the footing reinforcement to meet the minimum 0.12% requirement as per IS 456:2000, Cl. 26.5.2.1.

Development Length: Recalculate and specify correct development lengths for all bar diameters based on IS 456:2000, Cl. 26.2 and SP 34, Clause 43, ensuring 50d is not used as a blanket value if it falls short.

Concrete Grade: Clarify the environmental exposure conditions for the site in Kutch and confirm if M25 is adequate or if a higher grade (e.g., M30 or M35) is required for foundations as per IS 456:2000 Table 5.

Drawing Clarity: Add specific individual floor heights, clarify the basis for cross-section area calculations, and provide details on steel curtailment.